

The **Model Training Pipeline Design** for **SatQuery AI** is an auditable, domain-adapted machine learning pipeline tailored for remote-sensing vision-language tasks 1, 2. Rather than training a single monolithic model or using an unadapted generalist Vision-Language Model (VLM), the training architecture builds a **modular bank of specialized Low-Rank Adaptation (LoRA) adapters** over a single, frozen 1–4B parameter VLM backbone 3-6.

1. Data Ingestion & Quality Curation

The data pipeline processes open-source training datasets through structured storage and automated quality filtering before model fine-tuning begins 7:

1. **reBEN Ingestion & Storage Format:**The primary training dataset, **BigEarthNet.txt** (~464,044 co-registered Sentinel-1 SAR and Sentinel-2 multispectral image pairs with 9.6M text annotations), is ingested via reBEN (rico-hdl conversion) into an **LMDB / .safetensors** store 7, 8. This enables high-throughput random-batch reads without filesystem bottlenecks 7.
2. **Geographic Train/Val/Test Split:**Training uses an official **geographic split** (rather than a naive grid-based split) to prevent spatial data leakage between training and validation sets 7, 9.
3. **ScoreRS-Style Quality-Scoring Filter:**Before fine-tuning, a lightweight quality-scoring filter ranks all training image-text pairs and **drops the bottom quartile of noisy, blurry, or poor-quality annotations** 39, 46, 111, 111.1. Filtering the training pool improves downstream model performance more than switching backbones 4, 9, 10.

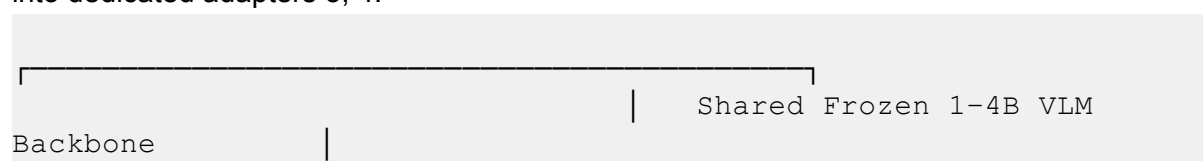
2. Backbone Architecture & LoRA Adaptation Strategy

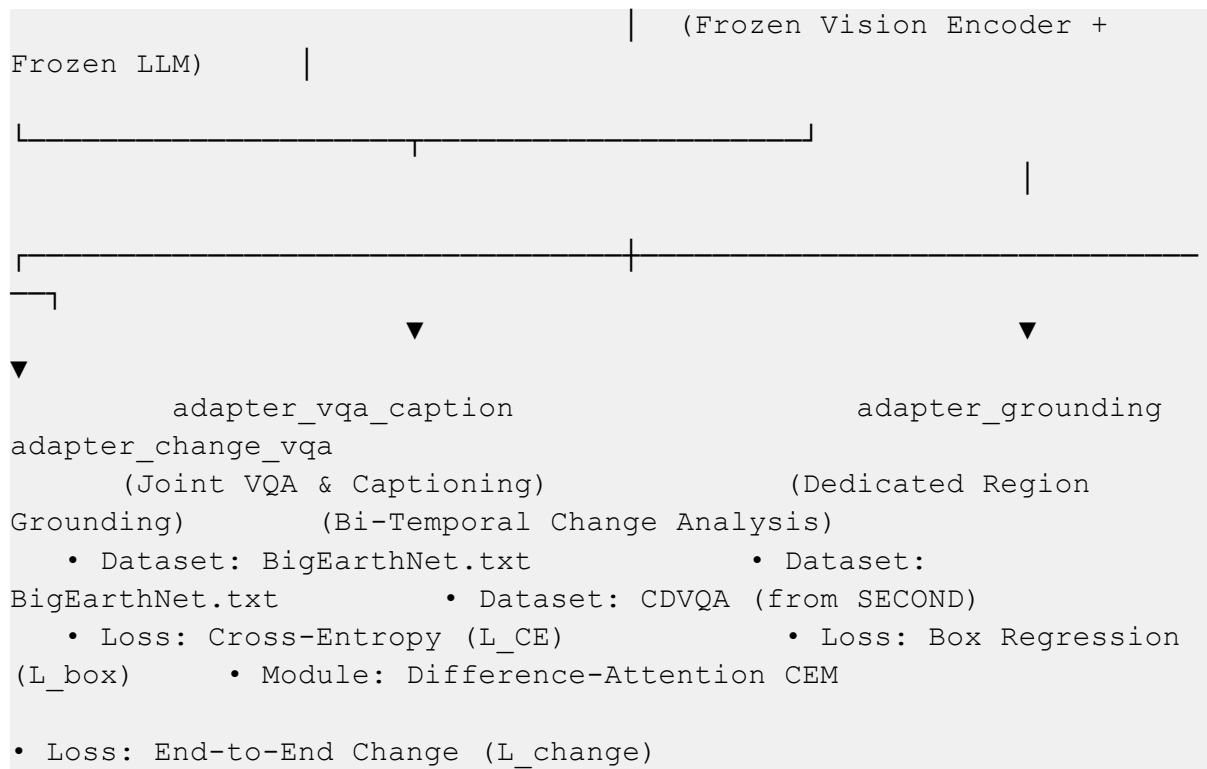
All specialist adapters share a single, compact **1–4B parameter open-weight VLM backbone** (e.g., **InternVL3-1B-class** or small **Qwen-VL variant**) 4:

- **Shared Frozen Backbone:** The heavy visual encoder (e.g., CLIP-ViT-L/14) and the base LLM decoder remain **completely frozen** 4, 11.
- **Modality Projection Bridges:** Small linear projection layers map Sentinel-1 (SAR) and Sentinel-2 (multispectral) tokens into the LLM decoder's embedding space 4, 11.
- **Low-Rank Adaptation (LoRA):** Trainable LoRA matrices (**rank 8–16**) are attached exclusively to the attention projections (Q, K, V, O) of the LLM decoder, updating **less than 1% of total model parameters** 4, 11.
- **Task-Conditioning Tokens:** Every fine-tuning sample prepends an explicit task token (e.g., [vqa], [caption], [ground], [change], [fusion]) to the instruction text to condition the backbone alongside adapter selection 12.

3. Specialist Adapter Training & Task Splits

To prevent task interference and **multi-task collapse** (where joint training across narrow datasets degrades specialist performance), SatQuery AI isolates task-specific knowledge into dedicated adapters 3, 4:





A. Joint VQA & Captioning Adapter (adapter_vqa_caption)

- **Training Data:** BigEarthNet.txt's captioning and binary/MCQ VQA annotations (filtered) 11.
- **Task Scope:** Open-ended scene descriptions, land-cover classification, and closed-set VQA 11, 13. For closed-set questions (e.g., yes/no or land-cover classes), decoding is constrained to the valid answer vocabulary 13.
- **Loss Function:** Autoregressive Token Cross-Entropy Loss ($\mathcal{L}_{\text{CE}}$) over target answer/caption token sequences 11: $\mathcal{L}_{\text{CE}} = -\sum_{t=1}^T \log P(w_t | \mathbf{x}_{\text{vis}}, \mathbf{x}_{\text{text}})$

B. Dedicated Region Grounding Adapter (adapter_grounding)

- **Training Data:** BigEarthNet.txt's referring-expression annotations (filtered) 11.
- **Reason for Task Isolation:** Grounding requires generating bounding-box token sequences 11, 14. Isolating grounding into its own adapter prevents its box-formatting constraints from interfering with natural-language text generation 4, 11.
- **Loss Function:** Sequence-to-Sequence Box Regression Loss ($\mathcal{L}_{\text{box}}$) 11. Spatial bounding box coordinates are tokenized into text strings ($[x_{\text{Left}}, y_{\text{Top}}, x_{\text{Right}}, y_{\text{Bottom}}, \theta]$), casting coordinate prediction as autoregressive token cross-entropy over coordinate tokens 11.

C. Bi-Temporal Change Analysis Adapter (adapter_change_vqa)

- **Training Data:** **CDVQA** (from the SECOND dataset) using the official train/val/test1/test2 split (retaining test2 as the distribution-shifted evaluation split) 15.

- **Architecture Addition:** Incorporates a **Change-Enhancing Module (CEM)** that applies difference-attention over Siamese-encoded (T_1) and (T_2) visual representations to highlight physical structural changes 15, 16.
- **Loss Function:** Difference-Attention End-to-End Change Loss $(\mathcal{L}_{\text{change}})$ 15. The CEM and text decoder are optimized end-to-end using token cross-entropy without auxiliary losses 15.

4. Optical-SAR Fusion Component Training

To avoid the performance degradation associated with naive early-fusion channel concatenation, optical-SAR cross-modal analysis uses a **two-stage structured fusion approach** 17-20:

- **Independent Evidence Extraction:** The single-image specialist extracts structured land-use/land-cover (LULC) evidence from optical and SAR images independently 18, 20.
- **Complementarity Detector Alignment:** A lightweight classifier trained on paired RGB-SAR data (**MM-OVSeg's CMU-Data** and **SOMA-1M**) identifies regions as agreement, optical_only, or sar_only 9, 18, 21.
- **Loss Function: InfoNCE Contrastive Alignment Loss** $(\mathcal{L}_{\text{InfoNCE}})$ aligns dense SAR-DINO and RGB-DINO feature representations across co-registered pairs 18, 21: $\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_{\text{sar}}, \mathbf{z}_{\text{rgb}}))}{\sum \exp(\text{sim}(\mathbf{z}_{\text{sar}}, \mathbf{z}_{\text{rgb}})) / \tau}$
- **Auditable Verbalization:** The LLM decoder converts the tagged evidence into a natural-language description, citing which sensor modality provided each claim 18, 22.

5. Domain-Gap Mitigation Pipeline

A core challenge in the training design is bridging the **5–30 $\times$ spatial resolution jump** and sensor mismatch between the training data (**Sentinel-1/2**, 10–60 m resolution) and the evaluation data (**Cartosat-2S optical**, 0.65–2 m resolution, and **RISAT SAR**) 18, 23-25:

- **GSD-Normalization Tiling:** Rescales spatial dimensions and tiles imagery to normalize pixel Ground Sample Distance 18, 25, 26.
- **Radiometric-Jitter Augmentation:** Applies histogram matching tailored to Cartosat-2S response curves and injects calibrated speckle noise matching RISAT SAR backscatter statistics 18, 25, 26.
- **Ablation Protocol:** Applied as an augmentation policy during training on a held-out slice to evaluate robustness under sensor domain shifts 18, 25.

6. Training Pipeline Summary

Component / Stage, Primary Dataset, Loss Function / Objective, Trainable Parameters

Data Filtering, "BigEarthNet.txt 7, 27, 28", "ScoreRS Quality-Scoring Model 39, 111.1", Upstream Filter 111.1

adapter_vqa_caption, BigEarthNet.txt VQA/Captions 11, Autoregressive Cross-Entropy $(\mathcal{L}_{\text{CE}})$ 11, LoRA $(r=8 \text{--} 16)$ + Projection 11

adapter_grounding, BigEarthNet.txt Referrings 11, Box Sequence Regression
 ($\mathcal{L}_{\text{box}}$) 11, LoRA ($r=8$) + Projection 11
 adapter_change_vqa, CDVQA (SECOND dataset) 15, End-to-End Difference Attention
 ($\mathcal{L}_{\text{change}}$) 15, LoRA + Change Module (CEM) 15
 Complementarity Detector, "CMU-Data + SOMA-1M 9, 21", "InfoNCE Contrastive Alignment
 ($\mathcal{L}_{\text{InfoNCE}}$) 18, 21", Dense Feature Encoder Pair 21
 Domain-Gap Policy, "Cartosat/RISAT Proxy Slice 9, 25", "Radiometric Jitter & GSD Rescaling
 18, 25, 26", Augmentation Policy 25

Compute Budget & Storage

Fine-tuning requires approximately **1–2 GPU-days per adapter** on a modern multi-GPU node 29. The trained LoRA deltas are stored as lightweight **.safetensors files (<50 MB per adapter)**, allowing all adapters to remain resident in GPU memory simultaneously for zero-reload hot-swapping during inference serving 6, 30.